

URBAN CHANGE DETECTION REPORT

AI/ML Based Urban Change Detection
using Drone and Satellite Imagery



Proof of Concept

BEFORE 2022
DRONE IMAGERY

AFTER 2025
HR SATELLITE IMAGERY



DRONE
IMAGERY



SATELLITE
IMAGERY



AI/ML BASED
ANALYSIS



GEOSPATIAL
INSIGHTS



ACCURATE CHANGE
DETECTION



DATE
7TH MAY 2026



PREPARED BY
AETOKSY TEAM

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1. Executive Summary

This report presents the results of an AI/ML-based Urban Change Detection Proof of Concept (POC) conducted over a 1 sq. km study area using multi-temporal drone and satellite imagery. The objective of the study was to identify, classify, and quantify urban changes between the year 2022 and 2025.

The analysis was performed using:

- Drone Orthophoto Imagery – Year 2022
- Open-source Satellite Imagery – Year 2025
- AI/ML-based Change Detection Workflow
- GIS-based Post Processing and Validation

The project successfully detected various categories of urban changes including new buildings, rooftop modifications, demolition activities, vegetation changes, road changes, and other urban development patterns.

The generated outputs include:

- Change Detection Polygon Shapefile
- Statistical Analysis Report
- Area-wise and Count-wise Change Analysis
- Visual Validation Screenshots

The results demonstrate the capability of AI/ML techniques for rapid urban monitoring, infrastructure assessment, and smart city planning applications.

2. Objective of the POC

The main objective of this Proof of Concept (POC) was to evaluate the effectiveness of AI/ML-based change detection techniques using heterogeneous geospatial datasets acquired from different time periods and platforms.

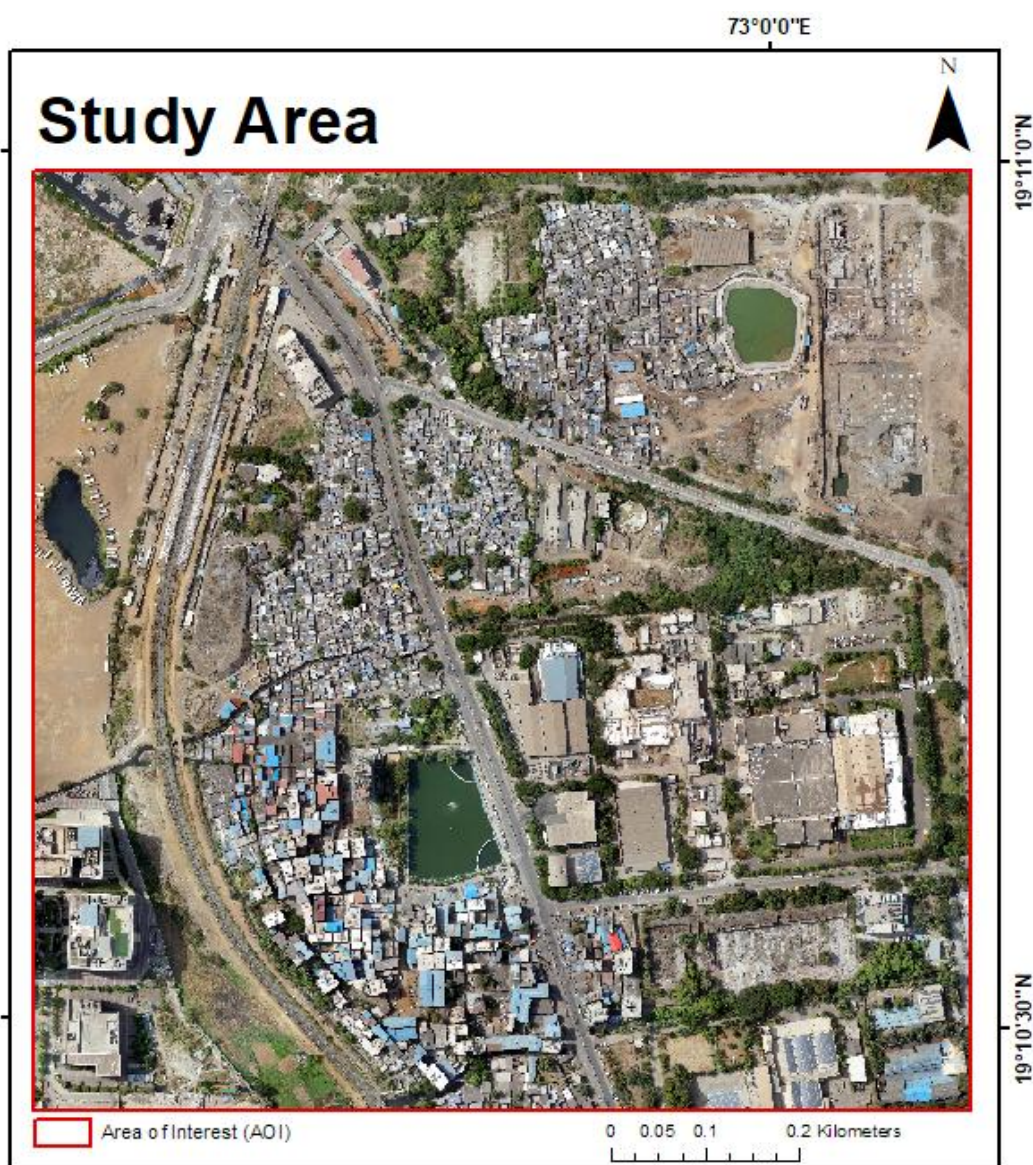
Specific objectives include:

- Detect urban changes between 2022 and 2025 imagery
- Classify detected changes into meaningful feature categories
- Quantify changes in terms of count and area
- Generate GIS-ready change polygons

3. Study Area

Parameter	Details
Study Area Size	1 Sq. Km
Area Type	Urban / Semi Urban
Analysis Type	AI/ML Urban Change Detection
Base Data Year	2022
Comparison Data Year	2025

Study Area Location Map



4. Input Datasets

4.1 Drone Orthophoto Imagery (2022)

Parameter	Details
Data Type	Orthophoto
Acquisition Year	2022
Platform	Drone/UAV
Usage	Baseline Reference Dataset

4.2 Satellite Imagery (2025)

Parameter	Details
Data Type	Satellite Imagery
Acquisition Year	2025
Source	Open-source Satellite Data
Usage	Current Year Comparison Dataset

5. Methodology

The AI/ML-based urban change detection workflow consisted of the following major stages:

5.1 Data Pre-processing

The input datasets were standardized prior to analysis.

Pre-processing activities included:

- Coordinate system normalization
- Orthorectification verification
- Spatial alignment and co-registration
- Resolution harmonization
- Image enhancement and normalization

5.2 AI/ML Change Detection Workflow

The change detection process was performed using AI/ML-driven feature extraction and comparison techniques.

The workflow included:

1. Feature Extraction from 2022 and 2025 datasets

2. Object-based segmentation
3. AI/ML-based classification
4. Temporal comparison analysis
5. Change polygon generation
6. GIS-based refinement and validation

5.3 Post Processing

Detected changes were refined using GIS-based spatial filtering and topology correction.

The following refinements were applied:

- Removal of false positives
- Elimination of noise polygons
- Feature refinement
- Attribute standardization
- Manual QA/QC verification

6. Change Detection Categories

The detected changes were classified into the following categories:

Sl. No.	Change Category	Description
1	New Building	Newly constructed buildings detected in 2025 imagery
2	Existing Building Extension	Expansion or extension of existing structures
3	Rooftop Change	Roof modification or rooftop alteration
4	Building Demolition	Removed or demolished structures
5	Vegetation Change	Significant vegetation increase/decrease
6	Road Change	Newly developed or modified roads
7	Water Body Change	Changes in water bodies
8	Other Change	Miscellaneous urban feature changes
9	Under Construction	Ongoing construction activities

7. Statistical Analysis

7.1 Count-wise Change Detection Statistics

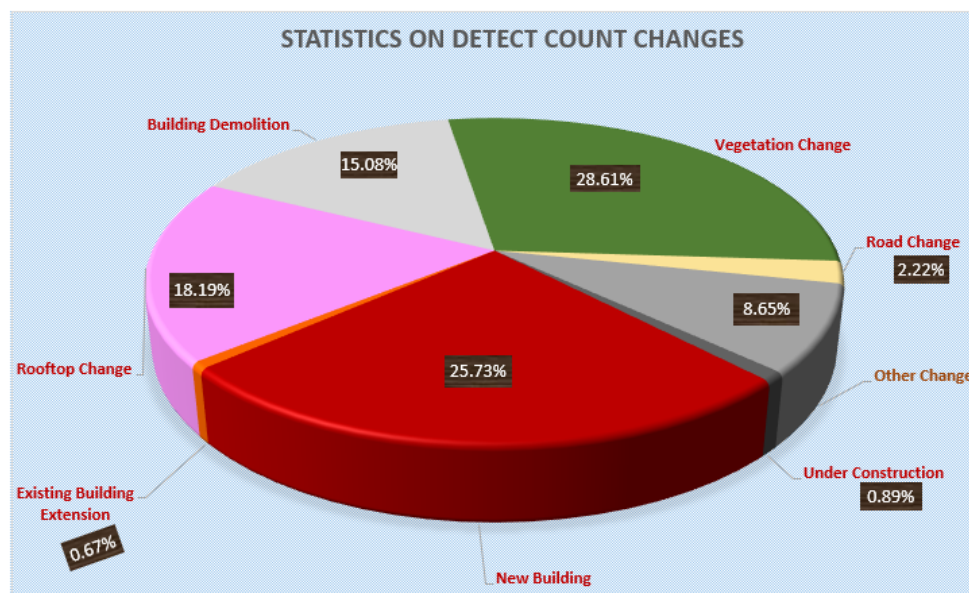
Detect Type	Detect Count	Percentage (%)
New Building	116	25.73

Existing Building Extension	3	0.67
Rooftop Change	82	18.19
Building Demolition	68	15.08
Vegetation Change	129	28.61
Road Change	10	2.22
Water Body Change	0	0.00
Other Change	39	8.65
Under Construction	4	0.89
Total	451	100.00

Key Observations

- Vegetation Change was the highest detected category contributing approximately 28.61% of the total detected changes.
- New Building detections contributed around 25.73%, indicating significant urban growth within the study area.
- Rooftop changes and building demolitions also represented major urban transformation activities.
- Water body changes were not observed during the analysis period.

Count-wise Change Detection Chart



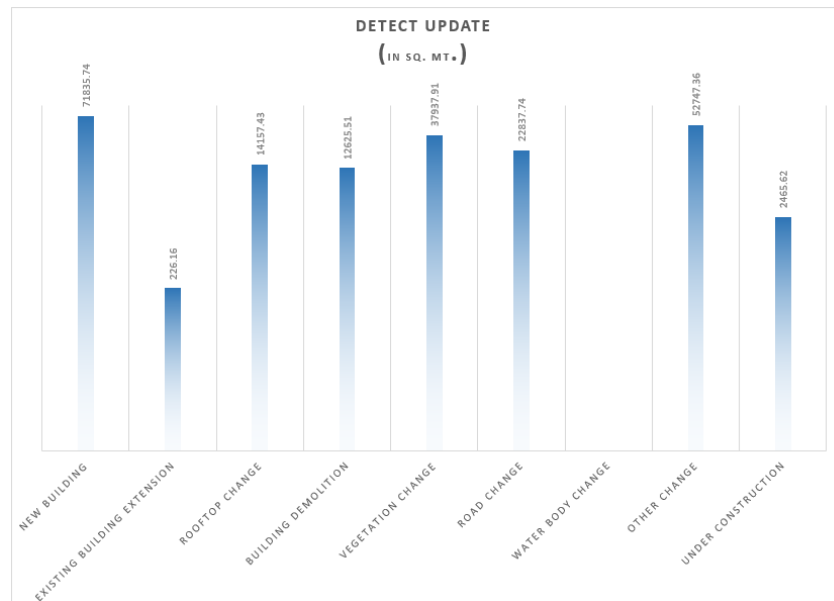
7.2 Area-wise Change Statistics

Feature Type	Percentage (%)	Area in Sq. Meter
New Building	33.44	71835.74
Existing Building Extension	0.11	226.16
Rooftop Change	6.59	14157.43
Building Demolition	5.88	12625.51
Vegetation Change	17.66	37937.91
Road Change	10.63	22837.74
Water Body Change	0.00	0.00
Other Change	24.55	52747.36
Under Construction	1.15	2465.62
Total Area	100.00	214833.47

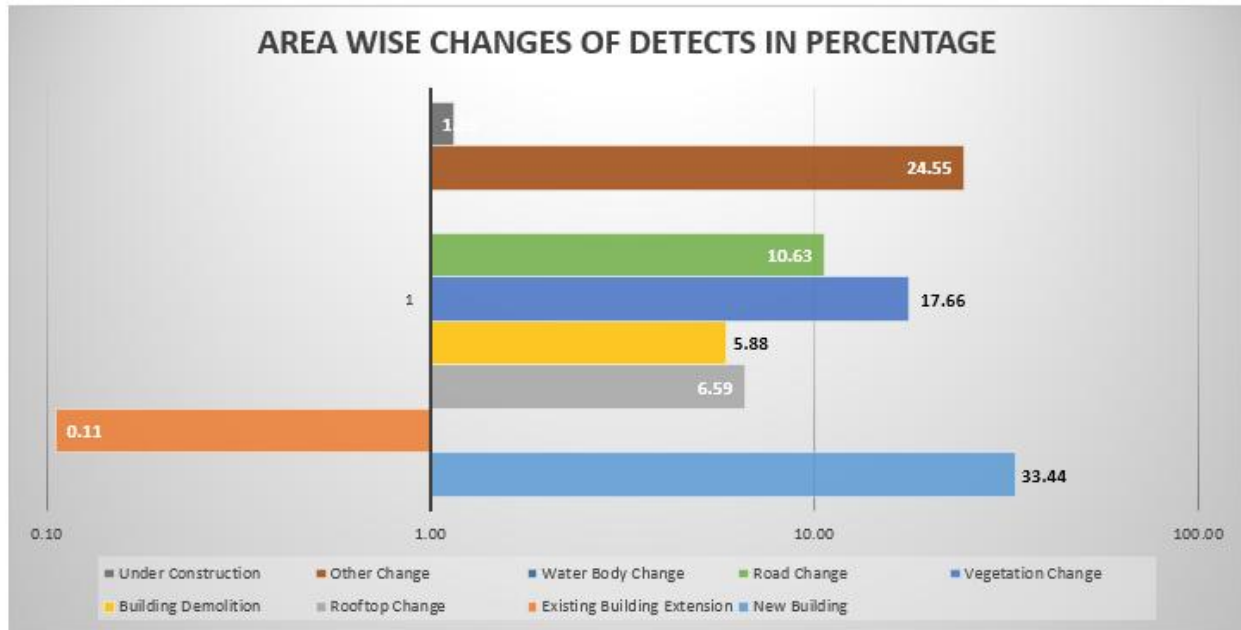
Key Observations

- New Building changes occupied the highest area coverage with approximately 33.44% of the total changed area.
- Other Change and Vegetation Change categories also showed significant spatial impact.
- Road Change contributed noticeably to urban infrastructure development.
- Under Construction activities were limited but spatially identifiable.

Area-wise Change Detection Chart



Area Statistics Visualization



8. AI/ML Detection Results

The AI/ML-based workflow successfully identified urban changes with high spatial consistency and clear feature separation.

Major Findings

8.1 Urban Expansion

The study area exhibited noticeable urban expansion primarily through:

- Newly developed building structures
- Building extensions
- Construction activities
- Road network modifications

8.2 Vegetation Dynamics

Vegetation-related changes were significantly detected across the study area, indicating:

- Land cover transformation

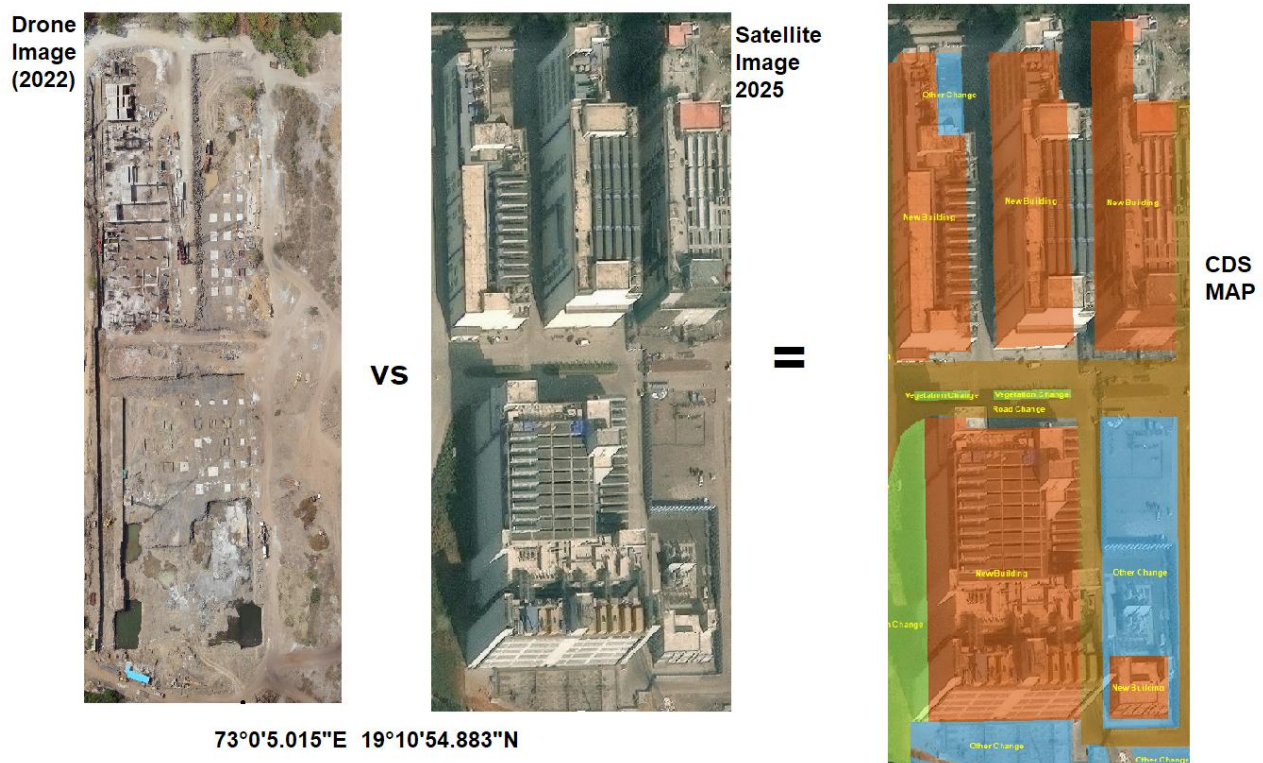
- Vegetation clearing activities
- Urban landscaping modifications
- Seasonal vegetation variation

8.3 Infrastructure Development

Road and construction-related changes indicate active infrastructure development within the study area.

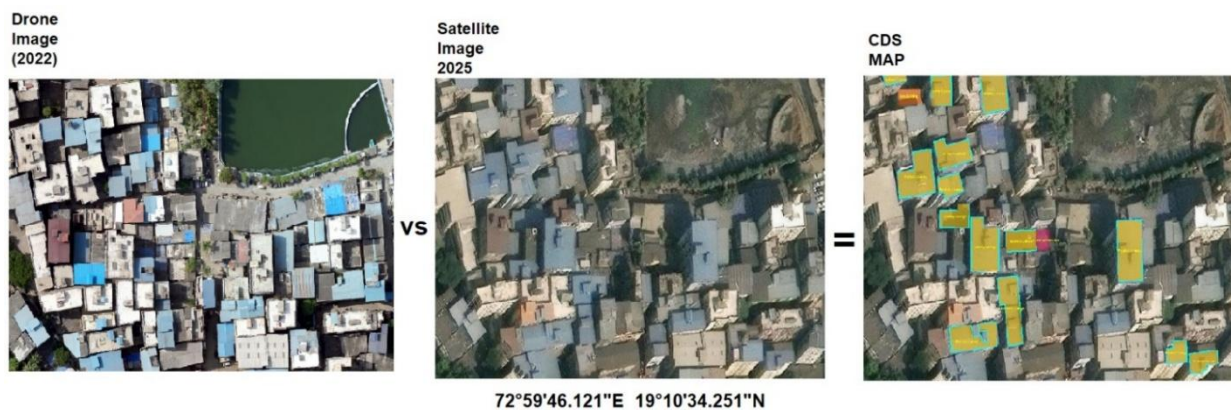
9. Sample Change Detection Outputs

9.1 New Building Detection





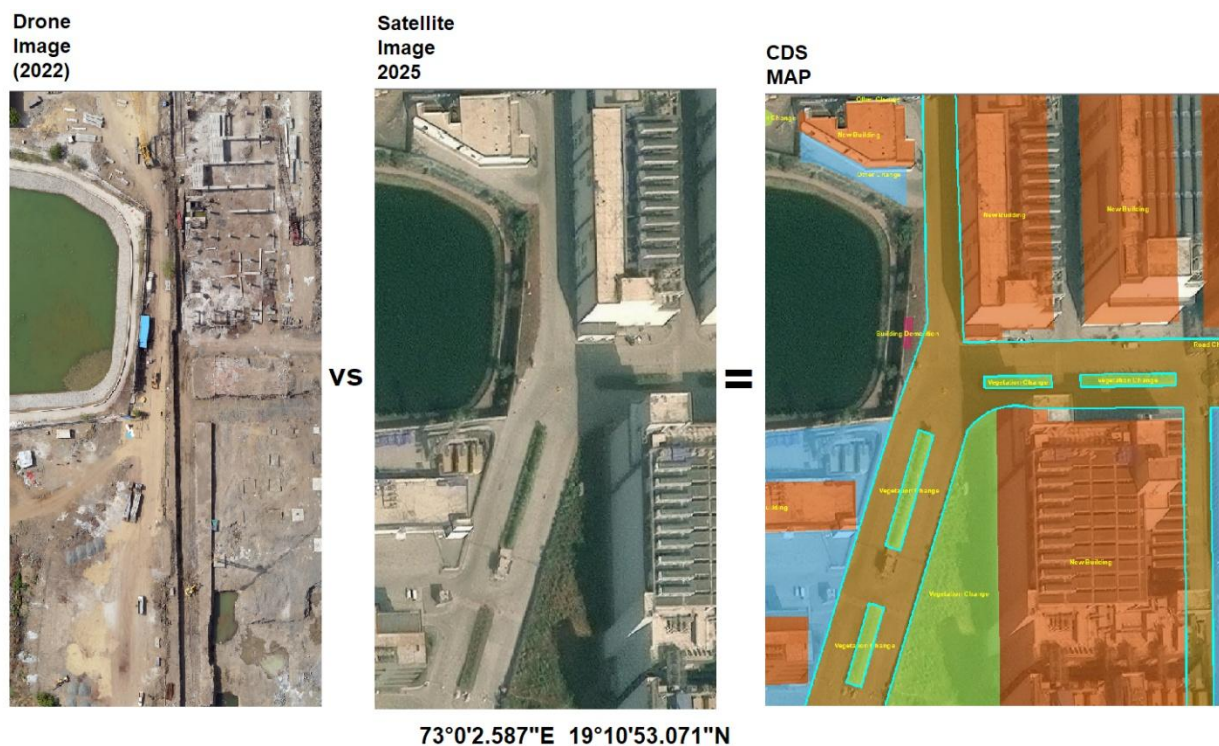
9.2 Rooftop Change Detection



9.3 Vegetation Change Detection



9.4 Road Change Detection



9.5 Demolition Detection



10. Accuracy and Validation

The detected changes were visually validated using GIS-based interpretation and manual quality checks.

Validation activities included:

- Visual comparison of 2022 and 2025 imagery
- Polygon boundary verification
- Attribute consistency checks
- False positive assessment
- Manual refinement of ambiguous regions

The AI/ML workflow demonstrated strong capability in identifying urban changes across heterogeneous imagery datasets.

11. Deliverables

The following deliverables were generated as part of the POC:

Sl. No.	Deliverable
1	Change Detection Polygon Shapefile
2	Statistical Report
3	Count-wise Change Analysis
4	Area-wise Change Analysis
5	Visualization Charts and Graphs
6	Sample Detection Screenshots

12. Conclusion

The AI/ML-based Urban Change Detection POC successfully demonstrated the capability of integrating drone orthophoto imagery and satellite imagery for automated urban monitoring.

The analysis identified a total of 451 urban change detections within the 1 sq. km study area. Major urban transformation patterns included new building development, vegetation change, rooftop modifications, and demolition activities.

The results confirm that AI/ML-driven change detection can significantly improve:

- Urban planning workflows
- Infrastructure monitoring
- Smart city development
- Land use monitoring
- Rapid change assessment
- GIS-based decision support systems

The workflow can further be scaled for district, city, or state-level operational implementation.